

System G: Fraud Detection System

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Roll#: 2K23/CSM/7

Domain: Digital Payments & Banking Transactions | Document Type: System Architecture Specification

Task 1: PEAS Specification

A structured evaluation of the Fraud Detection System framework utilizing the Performance, Environment, Actuators, and Sensors (PEAS) taxonomy:

PERFORMANCE MEASURES • Fraud Detection Rate (Sensitivity): Percentage of actual fraudulent transactions correctly flagged by the system. • False Positive Rate: Percentage of legitimate transactions incorrectly identified as fraud (minimized to protect client experience). • Processing Latency: Total inference and decisioning execution window (strictly targeted under 100 milliseconds). • Financial Loss Prevented: Cumulative monetary volume of fraudulent transactions successfully blocked.	ENVIRONMENT • Global interconnected digital banking architecture and APIs. • Online point-of-sale payment gateways and e-commerce nodes. • User endpoints, device telemetry profiles (IP addresses, physical locations, OS fingerprints). • Active transactional networks embedded with adaptive, highly sophisticated cyber-attack vectors.
ACTUATORS • Automated transaction execution and blocking APIs. • Real-time user notification engines (SMS, Email, Push Alerts). • Multi-Factor Authentication (MFA) step-up workflow engines. • Account freeze/suspension triggers propagated directly to core banking registers.	SENSORS • Streaming transactional ledgers (amounts, timestamps, Merchant Category Codes). • Historical consumer behavior and baseline profile logs. • Device context tokens (IP address, MAC address, session characteristics). • External threat intelligence parameters and central entity blacklists.

Task 2: Environment Classification

The operational environment of the transaction engine is classified across the standard dimensions of multi-agent cognitive architecture:

Dimension	Classification	Justification
Observability	Partially Observable	The system evaluates isolated footprints; it cannot explicitly access the true psychological intent of the user or unobserved peripheral components like compromised device roots or physical coercion.
Agents	Multi-agent	The architecture functions within an adversarial game theory environment against multiple distributed bad actors attempting fraud bypass models.
Determinism	Stochastic	State transitions are non-deterministic; client purchasing patterns undergo non-linear shifts and fraudulent methodologies mutate randomly without a certain causal relationship.
Episodic Nature	Sequential	Immediate transaction status adjustments alter future user profiles, baseline dynamic risk indices, system trust metrics, and sequential behavioral paths.
Dynamics	Dynamic	The multi-threaded environment continuously shifts at microsecond scale during inference, as millions of concurrent global transactions run while attackers actively evolve strategies.
Continuity	Continuous	Primary input elements consist of infinite numerical spectrums including fluid financial values, microsecond timestamps, and continuous geographic coordinates.

Task 3: Critical Analysis

1. Greatest Technical Challenge

The **Stochastic** properties of this operational matrix present the highest engineering hurdle for an AI platform. Because malicious networks constantly re-engineer their signatures to seamlessly mirror baseline user behaviors, feature-to-label boundaries are fluid and volatile. Engineering cannot resolve this through static heuristics or rigid threshold boundaries. It necessitates deep learning frameworks capable of processing subtle shifts and generalizing abstract vectors in real-time, preserving structural sensitivity while preventing an escalation of False Positives that would freeze standard financial movement.

2. Utility Function and Trade-off Behavior

Balancing optimal transactional security against customer service frictionless interactions requires an active multi-objective optimization utility model:

Proposed System Objective Function:

$$U = 0.7 \cdot \text{Accuracy} - 0.3 \cdot \text{False_Positive_Rate}$$

Impact of Parameter Tuning: If the mathematical coefficient corresponding to systemic security (Accuracy) is doubled relative to consumer convenience metrics, the algorithmic boundary defaults to a highly risk-averse operational profile. The agent will execute aggressive defensive patterns, intercepting transfers that manifest even minor mathematical deviations from nominal ranges. While this structurally maximizes risk mitigation and curtails fraud losses, it exponentially increases the False Positive Rate, culminating in significant customer friction due to false declines of authentic banking activities.

Task 4: Structured Representation (JSON)

The following standardized schema represents the semantic structure of the specification profile for system deployment integration:

```
{
  "system_name": "Fraud Detection System for Digital Payments",
  "peas": {
    "performance_measures": [
      "fraud_detection_rate",
      "false_positive_rate",
      "processing_latency",
      "financial_loss_prevented"
    ],
    "environment": "Global digital banking infrastructure, payment gateways, and evolving cyber-attack patterns.",
    "actuators": [
      "transaction_blocking_api",
      "mfa_challenge_trigger",
      "account_suspension_flag"
    ],
    "sensors": [
      "transaction_data_stream",
      "historical_user_logs",
      "device_fingerprinting_tokens"
    ]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "The system cannot know the true intent of the user or hidden external factors behind a transaction."
    },
    "agents": {
      "choice": "Multi-agent",
      "justification": "The AI agent operates against multiple malicious fraudsters actively trying to bypass detection."
    },
    "determinism": {
      "choice": "Stochastic",
      "justification": "Outcomes are unpredictable because user behaviors change and new fraud strategies emerge randomly."
    },
    "episodic_nature": {
      "choice": "Sequential",
      "justification": "Current flagging decisions affect future user trust scores and subsequent security states."
    },
    "dynamics": {
      "choice": "Dynamic",
      "justification": "The transaction environment continuously updates in real-time while fraud patterns adapt."
    }
  }
}
```

```
    },
    "continuity": {
      "choice": "Continuous",
      "justification": "Transaction values, precise timestamps, and geographic distances are continuous numerical quantities."
    }
  },
  "utility_function": "U = 0.7 * accuracy - 0.3 * false_positive_rate"
}
```